MUHAMMAD ABDULLAH

+92 328 4180034 | m.abdullah12589@gmail.com
GitHub: github.com/Muhammad-Abdullah-01
LinkedIn: linkedin.com/collab-with-muhammad-abdullah
Portfolio Website: muhammad-abdullah-portfolio-nu.vercel.app

EDUCATION

FAST-NUCES, LHR BS Computer Science

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Artificial Intelligence, Machine Learning, Object-Oriented Programming, Software Engineering

EXPERIENCE

Nolyth-AI — AI Engineer Intern *July 2026 – Present*

- Enrolled in a structured 4-sprint roadmap to AI Engineer; Sprint 1 completed.
- Built **New Life**, a full-stack habit-tracking app (FastAPI, Streamlit, SQLAlchemy/SQLite, JWT auth) with dynamic SVG analytics, streak tracking, and a GitHub-style contribution graph.

Optimus Automate — Web Development Intern *June – July 2026*

- Built **Converge**, a real-time chat application using the MERN stack with Socket.io, TanStack Query, and TailwindCSS, including role-based admin controls.
- Built **Buynest**, a production-style MERN e-commerce platform with JWT authentication, Stripe payments, Cloudinary media handling, and a full admin dashboard.

Decode Labs — Web Development Intern *June – July 2026*

- Built **TransitFlow**, a Python desktop bus reservation system with an interactive visual seating chart, Stripe-integrated ticket payments, and automated SMTP email receipts.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript (ES6+), SQL, x86 Assembly (NASM), HTML5, CSS3
Frontend: React.js, TailwindCSS, Framer Motion, Responsive UI Design, DOM Manipulation
Backend: Node.js, Express.js, FastAPI, Django, REST API Design & Development, SMTP Email Server Integration
AI/ML: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Feature Engineering, Prompt Engineering, Model Training & Evaluation, Data Preprocessing, Hyperparameter Tuning
NLP & LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), OpenAI API Integration, Hugging Face Transformers, LangChain, LlamaIndex, Transformer Models, BERT Fine-Tuning, Semantic Search
Databases: MongoDB, MySQL, SQLite, PostgreSQL, Database Design, Normalization, Query Optimization, Indexing, Schema Design
Tools: Git, GitHub, Vercel, Render, Docker (Basic Containerization), Postman API Testing, Linux CLI, Jupyter Notebook, Google Colab
Core: Machine Learning, Deep Learning, CNN Architectures, RNN/LSTM Models, Recommender Systems, System Design Basics, Software Engineering Principles, OOP, Data Structures & Algorithms, Computer Organization & Assembly Language (COAL)

PROJECTS

HealSync-AI (LLM + NLP System)

- Built an AI-powered medical chatbot for symptom analysis and preliminary health guidance, integrating the **OpenAI API** with a **LangChain** pipeline.
- Designed prompt-engineered conversational workflows to generate structured, context-aware, and reliable responses across user queries.
- Implemented a **TensorFlow**-based safety validation layer to detect and filter uncertain or unsafe outputs before delivering responses.

Tech Stack: Python, FastAPI, Streamlit, OpenAI API, SQLite

ScholarAssistant (AI Academic Research Assistant)

- Developed an AI assistant that performs semantic search queries across **arXiv** and **Semantic Scholar** databases.
- Built a PDF document chat feature implementing **Retrieval-Augmented Generation (RAG)** using the **OpenAI API**.
- Implemented citation recommendation and citation mapping, and evaluated search performance using NLP metrics (**ROUGE**, **BLEU**).

Tech Stack: Python, FastAPI, Streamlit, OpenAI API, SQLite

ProductPulse (AI-Powered Demand & Inventory System)

- Designed and built a full-stack dashboard featuring predictive demand forecasting and real-time inventory management.
- Implemented demand forecasting using a **linear regression** model (**scikit-learn**) alongside a **time-series model (ARIMA/Prophet)** to automate stock replenishment suggestions.
- Synchronized **MongoDB** updates dynamically across components for consistent inventory tracking.

Tech Stack: React.js, Node.js, Express.js, MongoDB, TailwindCSS, Python, scikit-learn, ARIMA/Prophet

Spotter Logbook AI (FMCSA-Compliant Electronic Logging Device App)

- Built a full-stack ELD web app that automatically generates **FMCSA**-compliant Hours of Service (**HOS**) logs.
- Implemented route planning with automated stop and rest-break calculation based on HOS regulations.
- Designed a React frontend backed by a **Django** REST API with a **PostgreSQL** data layer.

Tech Stack: React, Django, PostgreSQL, REST APIs

Sudoku Game (Low-Level x86 Assembly)

- Developed a fully playable Sudoku game in **x86 16-bit Assembly** running in DOSBox, with direct video memory rendering and custom keyboard input handling.
- Configured hardware timer interrupts for timer tracking and custom PC speaker sound for audio alerts.

Tech Stack: x86 Assembly, MASM, DOSBox

Klondike Solitaire Game (C++ Custom Implementation)

- Implemented a complete console-based Solitaire game with custom stack and doubly linked list classes to handle card state transitions and undo operations, without STL containers.
- Managed dynamic memory allocation and clean pointer cycles to prevent runtime memory leaks.

Tech Stack: C++, OOP, Custom Data Structures